8. Principles of Air Cleaning

It is necessary to dispose of the air that has been removed from the work environment. Often, depending upon the condition of the air exhausted, it is possible to introduce the contaminated air directly into the atmosphere through vents and stacks. In doing so, it is expected that the contaminant loading in the air will be sufficiently reduced by atmospheric dilution. However, in many cases the condition of the contaminated air is such that it must be treated before introduction into the atmosphere. The next two chapters will discuss in some detail the treatment of contaminated air prior to entry into the atmosphere.

The Reasons for Cleaning Air

1. Remove hazardous contaminants. One of the major reasons for an industrial exhaust system is to remove air containing potentially hazardous contaminants from the workplace. If the air is so contaminated that atmospheric dilution will not sufficiently reduce the contaminant level below that of a potential hazard, then the air must be treated in some manner before being vented into the atmosphere. Contaminants entering the atmosphere can cause a community hazard as well as a potential hazard for workers in plant areas adjacent to the contamination source. In determining the need for cleaning exhaust air, one should consider the possibility of an atmospheric inversion occurring which would limit the potential dilution rate that might normally occur. The air should enter the atmosphere in such a condition that under the worst possible case no potential hazard is present for those outside the facility.

2. Remove nuisance contaminants. Often, though no potentially hazardous contaminant exists in the exhausted air, a public nuisance may be created by the odor or amount of smoke present. The existence of such nuisance substances can cause community complaints and may detrimentally affect the public relations of the firm. Thus, although there may be no potential hazard from the air exhausted into the atmosphere, it may be necessary to clean the air of odors and smoke. This is particularly the case in areas where strict smoke control laws have been enacted.

3. Protect air-moving equipment. If the contamination in the air removed from the industrial environment contains abrasive or corrosive materials, it may be necessary to remove these materials from the air before passing it through the air-moving equipment (fans). If these contaminants are not removed, they can cause excessive wear from corrosion and abrasion on the air-moving equipment, thus necessitating costly repairs or frequent replacement.

4. Recover valuable materials. In some cases, the material contaminating the air has recovery value. If sufficient quantities of expensive materials are exhausted from the workroom atmosphere, the cost of air-cleaning equipment to remove these contaminants from the air may be partially or totally offset by the value of the materials that are recovered. These materials can then be recycled back through the production system.

5. Enable recirculation. Another reason for cleaning the air is to enable the recirculation of air removed from the work environment to other areas. Recirculation reduces the necessity for tempering the air (particularly heating in the winter). Although in the past recirculation was not widely used because of the problems inherent in cleaning the air and the potential for failure of the air-cleaning equipment (thus introducing a potentially hazardous contaminant into the work environment), more use of this method may be made in the future. As fuel costs rise, and the reliability and efficiency of equipment increases, recirculation may become feasible in terms of the savings generated.

6. Meet environmental requirements. One final reason for cleaning the exhaust air is to meet the environmental requirements in the area in which the plant is located. Environmental laws concerning air pollution have become stricter in the past few years, and it is likely that even more stringent laws will be introduced in the future. Thus, it may be necessary to install air-cleaning equipment in order to meet the requirements of environmental laws.

Factors Affecting Air Cleaning

When it becomes obvious that air-cleaning equipment is necessary in a particular industrial exhaust system, an engineering study must be conducted to determine the type and size of equipment to be installed. There are a number of factors that affect the type of equipment to be installed. These factors should be considered from the outset in order to implement the appropriate system. The major factors to be considered in such a study are:

a. Type of contaminant.
b. Size and shape of particulate contaminant.
c. The volume of air handled.
d. The degree of collection required.
e. The toxicity of the contaminant.
f. The radioactivity of the contaminant.
g. The heat and humidity of the carrier gas (air).
h. The corrosiveness of the contaminant material.
i. The presence of abrasive particulates.
j. The disposal methods for the collected contaminant.
k. The flammability of the carrier gas or the contaminant.
l. Chemical reactions that may occur.
m. Pressure loss in the cleaning mechanism.
n. The integrity of the air cleaner.
o. Variations in contaminant loading.
p. The necessary cleaning cycle.
q. The cost of cleaning the air.
r. The efficiency of the air cleaner to be installed.

The following discussion will briefly cover each of these factors.

The Type of Contaminant. The contaminant in the air may be either a particulate or a gas/vapor. Particulate matter includes dust, fumes, smoke, and mists. Different cleaning methods may be best for a given type of contaminant. For example, specific methods are most applicable for the removal of particulate matter, while other methods are more appropriate for the removal of gases and vapors. The Size and Shape of Particulate Contamination. The size and shape of a particle will cause it to move through the air and to be acted upon by the air cleaner in different ways. This subject was discussed in some detail earlier when the properties of airborne contaminants were considered. The shape of an individual particle will influence its movement in the exhaust system and cleaning mechanism. The density will also be a factor in such a system. Obviously, from previous discussions, the size of particle has a major influence on its inertial and gravitational attraction. Particles less than 10 micrometers generally follow the movement of the air and are not materially affected by gravity or the inertia of the individual particle. A general rule is that the smaller and less dense the particle, the higher the cost required to remove it from the air. Certain air-cleaning methods are practical only for large particles, while others may be used to remove particles in the submicrometer range.

The Volume of Air Handled. For a given concentration, the larger the air volume handled, the greater the need for better air-cleaning equipment. Obviously, if a small amount of air is being exhausted, even though this air may have a high concentration of contaminant, the total amount of contaminant entering the environment is small. On the other hand, if a large volume of air is being exhausted with only a small percentage of contaminant, the total contaminant entering the atmosphere is significantly large and requires efficient air-cleaning equipment.

The Degree of Collection Required. The degree of collection of contaminant required will vary depending on a number of factors. The environmental laws may state the amount of contaminant that can be exhausted into the air. The potential hazard to the community will also be a factor. The location of the plant and the public relations image that is desired may also be considered when determining the degree of collection that is required. Many other factors are also important in this decision. Generally, it can be stated that the higher the degree of collection required, the higher the cost of the necessary air-cleaning equipment.

The Toxicity of the Contaminant. Toxic contaminants in the exhausted air can present potential hazards to the neighboring property and community. The presence of an atmospheric inversion may result in a potential hazard even though small quantities of the contaminant are released and normally diluted by air movement in the atmosphere. Generally, a toxic contaminant requires that the air cleaner efficiency be high. Other important factors to consider when working with toxic contaminants are the problems involved in disposing of the collected contaminant, and the special care that is required in the

maintenance of the air cleaner and the duct leading to the air cleaner to remove any potential hazard to the maintenance worker.

Radioactivity of the Contaminant. Radioactive contaminants are highly toxic materials. Thus, they should be treated with the same care as is necessary for other toxic material. Additional precautions must be taken to assure that radioactive buildup does not occur within the air cleaner, resulting in a potential exposure of the workers.

Temperature and Humidity. In many cases, air removed from the workplace will be at a high temperature or may contain a high moisture content. These factors can affect the operation of certain types of air cleaners and must be considered when selecting the appropriate cleaner. High humidity can cause particulate matter to agglomerate and plug narrow sections of the air cleaner. Humidity in the exhausted air also may cause a vapor plume from the stacks which results in a possible community problem. The moisture in the air can also result in corrosion of the air-cleaning and air-moving equipment unless appropriate protective precautions are taken.

The Presence of Corrosive Contaminant Material. The air exhausted from the workplace may contain materials that are corrosive. These materials can affect the maintenance of the air-moving equipment. As such, it may be desirable to remove the materials before passing the air through the air mover. In addition, special consideration must be given to the type of air cleaner, since corrosive material can attack the cleaner and cause excessive wear.

The Presence of Abrasive Particulates. As in the case of corrosive contaminants, the presence of abrasive particulates in the air will cause damage to the air-moving and air-cleaning equipment. The type of equipment to be used must be specially designed to protect from this type of wear. In addition, abrasive particles can cause a buildup of dust within the system unless the appropriate transport velocities are present to prevent such a buildup.

Disposal of the Collected Contaminant. It is important to consider the methods of disposing of the collected contaminant during the design of the system. If the contaminant is toxic, it may present a danger to workers maintaining and/or cleaning the equipment in the exhaust system. In addition, if water is used to collect the contaminant, the engineer must consider how this water will be disposed of without causing a community water pollution problem. If the potential salvage value of the contaminant is high, a method must be devised to recover the contaminant for reuse.

Flammability of the Carrier Gas or the Contaminant. If the carrier gas or the contaminant is flammable, certain precautions must be taken. The concentration of the flammable material must be kept above the upper explosive limit (UEL) or below the lower explosive limit (LEL) to remove the potential for a fire or an explosion. In addition, the cleaning equipment and air-moving equipment must be designed to eliminate that potential for sparks which may ignite the flammable material.

Chemical Reactions. The carrier gas and the contaminant may react with each other, thus resulting in a material different from that which was originally produced. In addition, the contaminant may react with water vapor in the air, or the carrier gas may react with materials present in the cleaning mechanism. The potential for such reactions must be considered, since they may have a major effect on the type of cleaning that is necessary.

Pressure Losses. When air is passed through a cleaning device, it loses pressure because of turbulence and friction. These losses will be reflected in the static pressure of the system. Depending on the type of air-cleaning device used, the pressure loss will vary from a very small loss (one-half inch water gauge) to a very large pressure loss (20 inches water gauge or more). As the pressure loss in the system increases, the required size for the air-moving equipment will also increase. This will result in a higher initial cost for installation of air-moving equipment and also a higher operating cost for the system. These factors must be considered during the design of the air-cleaning system.

The Integrity of the Air Cleaner. The air-cleaning mechanism must be such that leaks do not occur. If leaks occur on the supply side of the fan, these leaks can cause the contaminant to be dispersed in the atmosphere. On the exhaust side of the fan, leaks in the air cleaner and duct will require larger volumes of air to be exhausted, thus requiring a larger air mover. In addition, there is a potential for re-entry of the contaminant that has been collected unless precautions are taken to prevent leakage between the cleaner and the collection hopper.

Variations in Contaminant Loading. Variations in the contaminant loading of the air may have an effect on the air-cleaning mechanism. Low loading may affect the efficiency of certain types of air cleaners. On the other hand, high loading may require a high frequency of cleaning for the air cleaner.

The Cycle for Cleaning. The method for cleaning the air cleaner must be considered. If the operation is continuous, it may be necessary to operate the air cleaner constantly. This will require either the use of redundant air cleaners or implementing some type of continuous cleaning mechanism so that the process does not need to be shut down to clean the air-cleaning mechanism.

The Cost of Cleaning the Air. When comparing methods of air cleaning, it is necessary to determine the costs that are incurred using various methods in order to choose the one that is the most cost effective. Three major factors are important in such a cost comparison:

a. The initial installation cost of the air-cleaning equipment.
b. The cost of the power necessary to operate the system (including the cost of the power to operate the air-moving equipment necessary to overcome pressure losses occurring within the air cleaner).
c. The cost of maintenance and cleaning of the system.

Each of these factors must be considered in detail to determine the overall cost effectiveness of each system under consideration. It may be that to accomplish the job only one system can be used. However, even in this

case, it is desirable to determine the cost for operating the system to be used.

The Efficiency of the Air Cleaner. The type of air cleaner that is used will yield different results, depending upon the type of contaminant and the size of the particles present in the contaminant. A common method for stating the efficiency rating of a given air cleaner is

$$\text{Efficiency} = 1 - \frac{\text{Contaminant discharge rate at outlet}}{\text{Contaminant discharge rate at inlet}} \qquad (2.8.1)$$

Care must be exercised in the use of this expression since it is desirable to consider the total contaminant that is exhausted into the atmosphere rather than the percent or ratio that is volume-time related. For example, consider an air cleaner that is rated at a 90 percent efficiency and operates to remove silica dust that is generated at a rate of 12,000 pounds per hour. If the above expression is used, the following is obtained:

$$\text{Efficiency} = 1 - \frac{\text{Contaminant discharge rate at outlet}}{\text{Contaminant discharge rate at inlet}}$$

$$0.90 = 1 - \frac{x}{12{,}000}$$

$$x = 1200 \text{ pounds per hour}$$

Thus at a 90 percent efficiency, a large quantity of dust is still being exhausted into the atmosphere.

The effect of air cleaning on the quality of the vented air can be obtained in a number of ways. Dust samples and counts may be taken at the inlet and outlet of the air cleaner. The opacity of the plume at the stack may be compared to existing standards. The weight of solids gathered per unit of air handled may be obtained. Other standard air-quality tests may also be used to determine the air-cleaner efficiency.

Characteristics of the Contaminant

Particulate contaminants include dust, fume, mist, and smoke. Small particles will not settle from the air (particles that are of a diameter of less than 30 to 40 micrometers). Large particles, on the other hand, will settle from the air unless the transport velocity of the system overcomes the gravitational attraction of these particles. The particles of interest to the industrial hygiene engineer because of their respirability are those with a diameter of less than 10 micrometers. The particle size in any given air sample is distributed log-normally with most of the particles in the smaller range. The problem of air cleaning, then, is to remove this large number of small particles.

On the other hand, the problem may involve a gas or vapor. The gas or vapor is mixed with the air and does not settle. Some method must be present

to remove this gas or vapor from the air which does not involve gravitational attraction.

In general, these two categories of contaminants (particulate and gas/vapor) will result in two different methods of air cleaning. In the following discussions, the general methods of air cleaning will be considered. Not all of these methods will be equally appropriate for particulates or gases and vapors.

General Methods of Air Cleaning

The basic law of air cleaning can be stated as follows:

> The contaminant must be acted upon by a force which does not act on the carrier gas or acts with sufficient differential force to physically or chemically remove the contaminant.

The force must be great enough to overcome other forces within the air stream. If a particle exists in an air stream, it must be acted upon by a force that will cause it to travel across the air stream. The stream can be narrowed to decrease this distance of travel. In the following discussion, the basic methods for cleaning will be presented. The various types of air cleaners that are presented in the next chapter use one or more of these basic methods as the means for removing contaminants from the air. The basic methods of air cleaning are:

a. Gravitational force
b. Centrifugal force
c. Inertial impaction
d. Direct interception
e. Diffusion
f. Electrostatic precipitation
g. Adsorption
h. Absorption
i. Incineration
j. Catalytic combustion

Gravitational Force. As particles travel in the air stream, they are acted upon by gravitational force. If the force of gravity is larger than the force exerted on the particles to transport them with the air stream, the larger, more dense particles will settle from the air. This approach for cleaning air is generally useful only with particles that a-e large and dense. As has been previously discussed, small particles have a very slow terminal settling velocity; the force necessary to transport them is relatively small.

Centrifugal Force. If air is caused to rotate around a small diameter, a force much greater than that of gravity will act on the particles to remove them in a direction tangential to the direction of rotation. Thus, the particles will be projected across the rotating stream of air. If a barrier exists to halt the particles' tangential motion, these particles can then be captured on the outer edge of the rotating air stream. Such a method will

operate to capture particles much smaller than those that are normally affected by gravitational force.

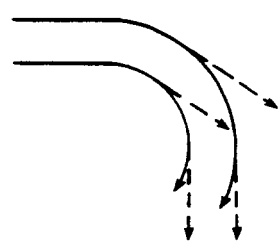
Figure 2.8.1

Gravitational attraction.



Figure 2.8.2

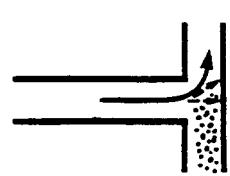
Centrifugal force.



Inertial Impaction. As air flows, it diverges in order to pass around objects present in the air stream. The inertia of each particle in the air stream causes these particles to flow in a more nearly straight line. The particles may then impact on the deflecting object. This method is more effective for large particles that have an inertia apart from that which is imparted by the air stream. In general, the higher the air velocity, the higher the probability of impaction of particulate matter.

Figure 2.8.3

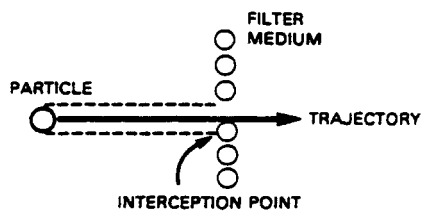
Inertial impaction.



Direct Interception. As air passes around an object or series of objects that are placed in the air (as in the case of inertial impaction), the particles may strike or be intercepted by the object(s). The finite width of the particle may cause the particle to be intercepted even though the trajectory calculated at the center of the particle may pass through the object. Thus, the larger the particle or the smaller the distance between the objects, the greater the probability of interception. Interception is independent of the velocity of the air.

Figure 2.8.4

Direct interception.

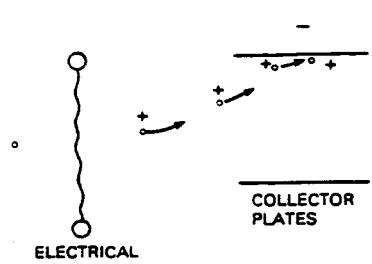


Diffusion. Small particles (those less than 0.5 micrometers) are caused to separate from the air by Brownian motion (caused by molecular impact on the small particle). The diffusion process is a slow process and requires the air to remain in the area where cleaning takes place for a period of time. In addition, low velocity air movement is required. Often, a filter medium is used that then combines interception methods with diffusion methods.

Electrostatic Precipitation. In this method, the particle is given a charge as it flows through an electrical field. A collector plate is given an opposite charge. As the particle passes by the collector plate, it is attracted to the oppositely charged plate. Some particles may have a natural charge and, thus, do not require passing through an electrical field.

Figure 2.8.5

Electrostatic precipitation.



Adsorption. In the process of adsorption, a contaminant in the air passes through an adsorbing material. The molecules of the contaminant adhere to the surface of the adsorption medium. Adsorption is basically a physical process. The adsorbate-adsorbent relationship is generally a specific one with certain materials acting as adsorbents for other materials.

Absorption. In absorption, a soluble or chemically treated reactive contaminant in the air is caused to flow through water or a chemical medium. The contaminant is dissolved in the liquid or reacts with the medium in such a manner that its form is changed, and it is removed from the air.

Incineration. The contaminant-containing air is passed through an open flame. The combustible contaminant then burns off, producing only carbon dioxide and water.

Catalytic Combustion. In catalytic combustion, a catalyst is introduced to the contaminated air. This catalyst causes an oxidation of the contaminant to take place. The presence of the catalyst causes this oxidation to take place at a lower temperature than would occur in incineration, and no flame is present during the oxidation.

Summary

Among the many reasons for cleaning air exhausted from the workplace, perhaps the most important is the removal of potentially hazardous contaminants before they enter the atmosphere. Before selecting the air cleaner to be used, the industrial hygiene engineer must be familiar with the process and the contaminant of concern. Various methods of air cleaning are available, but not all are equally effective on all contaminants. The most common principles used in air cleaners are gravitational forces, centrifugal force, inertial impaction, direct interception, diffusion, electrostatic precipitation, adsorption, absorption, incineration, and catalytic combustion. It is not uncommon to find more than one of these principles utilized in a particular air cleaner design.